

Challenge (Habanero)

- Q1.** A chef's oven must stay within $12^\circ F$ of $375^\circ F$. If T is the actual temperature, which absolute value inequality represents the tolerance?
- (A) $|T + 375| \leq 12$
(B) $|T - 375| \leq 12$
(C) $|T - 12| \leq 375$
(D) $|375 - 12| \leq T$
(E) $|T| - 375 \leq 12$
- Q2.** Evaluate the expression for a structural beam's alignment error: $|-14 + 5| - |8 - 15|$.
- (A) -16
(B) 2
(C) 16
(D) 0
(E) -2
- Q3.** A construction site uses a vertical axis where ground level is 0. A foundation pile is at -22 feet and a crane hook is at $+45$ feet. What is the total distance between them?
- (A) 23 feet
(B) -23 feet
(C) 67 feet
(D) $|-22| - |45|$ feet
(E) 45 feet
- Q4.** A pastry recipe calls for x grams of sugar. The absolute error is defined as $e = |x - 120|$. If the chef uses 112.5 grams, find the value of e .
- (A) -7.5
(B) 120
(C) 7.5
(D) 232.5
(E) 8.5
- Q5.** Find the sum of all possible values of k if the distance between a support pillar at k and a reference point at 5 on a blueprint is exactly 13 units.
- (A) 18
(B) -8
(C) 10
(D) 13
(E) 26
- Q6.** Simplify the measurement expression: $-|-18.4| + |-2.1 - 3.9|$.
- (A) -12.4
(B) -24.4
(C) 12.4
(D) -14.4
(E) 24.4
- Q7.** In a climate-controlled cellar, the temperature deviation is $|x - 55| < 3$. Which temperature is NOT within this construction specification?
- (A) $53.1^\circ F$
(B) $57.9^\circ F$
(C) $55.0^\circ F$
(D) $52.0^\circ F$
(E) $56.5^\circ F$
- Q8.** Evaluate the height difference between two architectural points A and B , where $A = |-12 \cdot 3|$ and $B = -|-5^2|$.
- (A) 11
(B) 61
(C) -11
(D) 36
(E) 1

Open Ended Questions (FRQ)

- Q9.** An architect designs a ramp where the start is at elevation -4.5 meters and the end is at $+2.3$ meters. Use an absolute value expression to calculate the total vertical rise and solve.
- Q10.** A high-precision cooking thermometer has a calibration error defined by $|C - A| = 0.8$, where C is the displayed temp and A is the actual temp. If the display shows $212.0^\circ F$, calculate both possible actual temperatures.

Chef's Tips

- Ensure students visualize absolute value as distance on a number line.
- Verify LaTeX rendering for all magnitude bars.
- Check that students differentiate between $-|x|$ and $|-x|$.